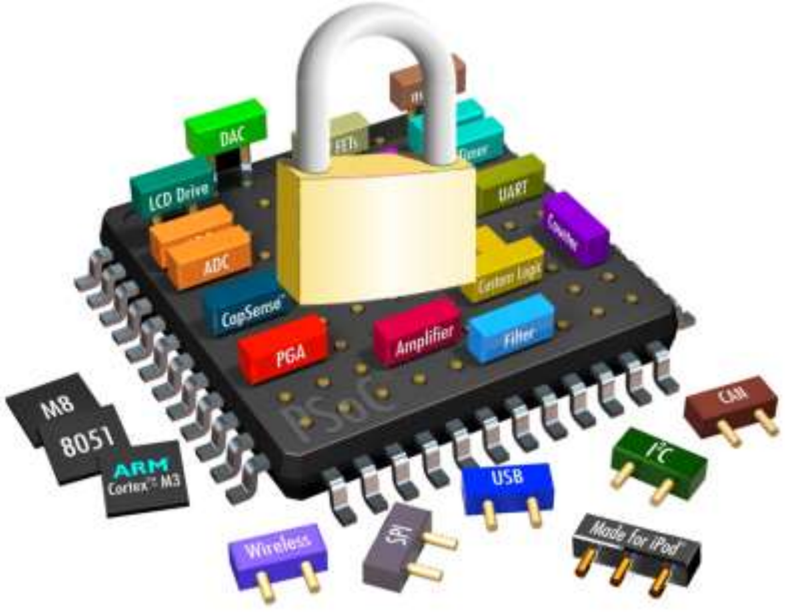
Hardware Security

Intel SGX: Architecture and Security Vulnerabilities



SGX Security Objectives

01

Integrity Protection

SGX must detect any integrity violations of enclave code or data, preventing access to modified content and ensuring execution authenticity.

02

Confidentiality Guarantee

Security-critical code and data within enclaves must remain confidential, protected from unauthorized observation or extraction.

03

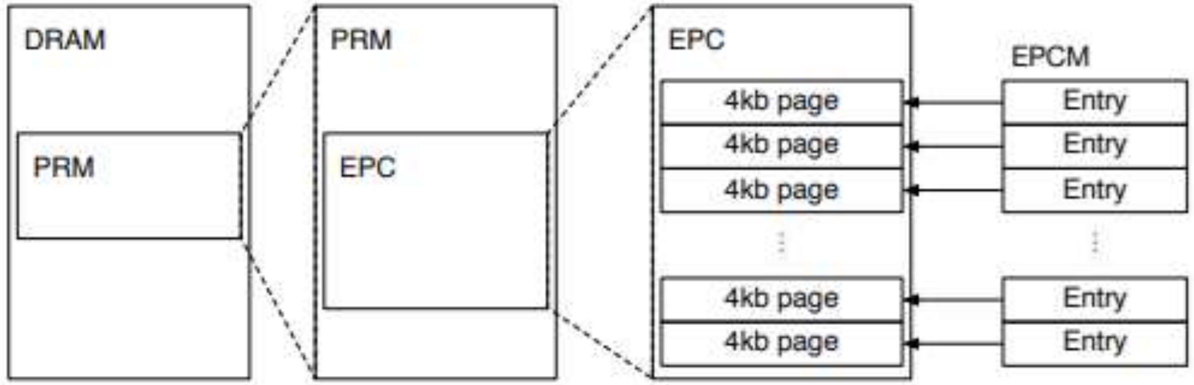
Isolation Assurance

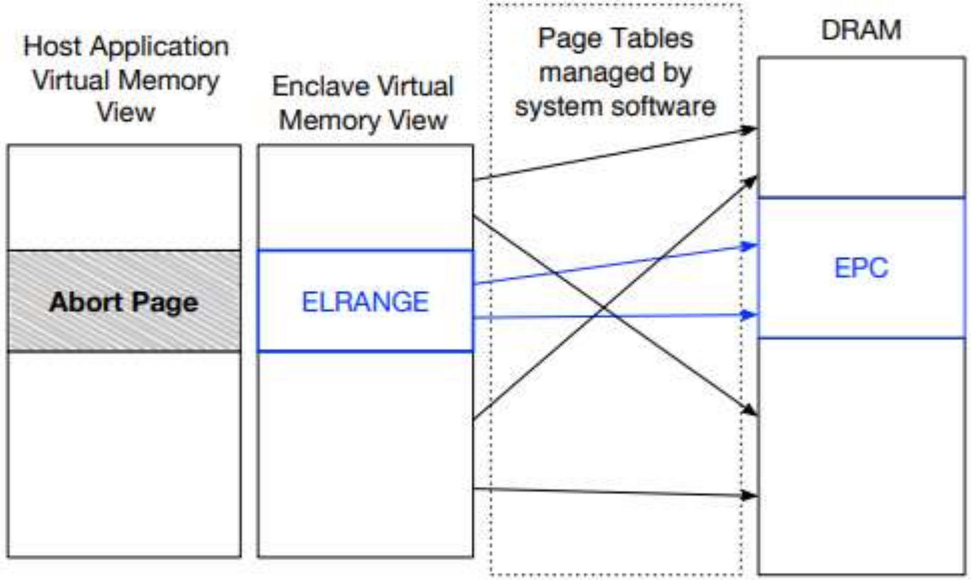
Multiple enclave instances must coexist securely on the same platform with guaranteed isolation between different enclaves.

SGX Execution Model

- SGX applications execute in two distinct regions: a normal untrusted region and a security-critical enclave region.
- The host process initiates an enclave, loads sensitive code and data into protected memory, and calls enclave functions when security-critical operations are needed.
- Only code executing within the enclave can access Processor Reserved Memory (PRM).
- After enclave functions complete, results return to the host process while enclave contents remain protected in isolated memory space.

Memory Organization in SGX





01

Processor Reserved Memory (PRM)

A protected memory region that the CPU shields from all non-enclave accesses, including kernel, hypervisor, SMM, and DMA transfers from peripherals.

02

Enclave Page Cache (EPC)

4KB pages within the PRM that store enclave code and data. System software manages EPC allocation while the CPU tracks each page's state.

03

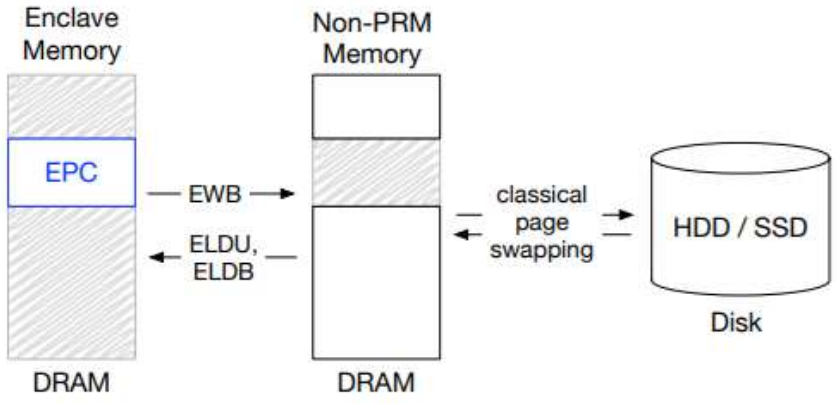
Enclave Page Cache Map (EPCM)

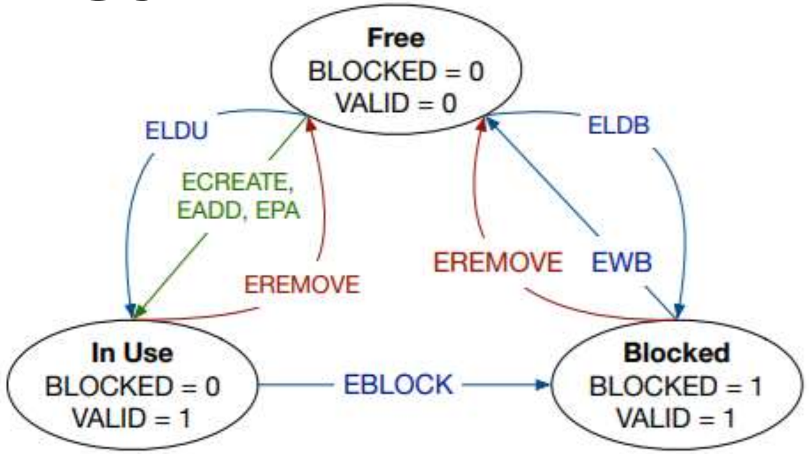
The Enclave Page Cache Map ensures each EPC page belongs to exactly one enclave, preventing unauthorized access and maintaining isolation.

Critical Data Structures

- SECS
  - SGX Enclave Control Structure stores enclave metadata in dedicated EPC pages, accessible only by CPU hardware for enclave management operations.
- TCS
  - Thread Control Structure enables concurrent execution within enclaves, with each TCS stored in a dedicated EPC page for thread management.
- SSA
  - State Save Area securely stores enclave execution context during hardware exceptions, protecting sensitive state from untrusted system software.

EPC Page Eviction Strategy





EBLOCK

Marks pages for eviction and prevents new TLB entries from being created for blocked pages.

ETRACK

Tracks which logical processors need TLB flushes before eviction can safely proceed.

EWB

Encrypts and MACs the page contents, generates a nonce, and writes everything to untrusted DRAM.

ELDU/ELDB

Loads evicted pages back into EPC, verifying integrity and freshness before restoring.

Address Translation Security

Virtual Memory Isolation

Each enclave designates an ELRANGE in its virtual address space for EPC pages. The CPU enforces that enclave code can only access memory within this range or explicitly shared non-EPC memory.

Page Table Verification

While untrusted system software manages page tables, SGX verifies every address translation to ensure it matches the enclave author's intended memory layout and access permissions.

Attack Prevention

SGX's checks prevent malicious OS from performing memory mapping attacks, such as remapping enclave pages to different virtual addresses or modifying access permissions.